

Name: Key

Date:

Topic:

Class:

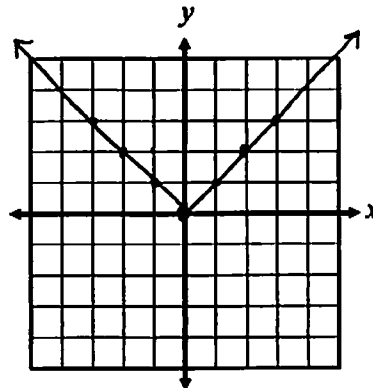
Main Ideas/Questions

Notes/Examples

ABSOLUTE VALUE
FUNCTIONS

- The **absolute value function** is written as $f(x) = |x|$
- Complete the table below to graph the function. Give the domain and range.

x	$ x $
-3	3
-2	2
-1	1
0	0
1	1
2	2
3	3



$$D = \{ \mathbb{R} \}$$

$$R = \{ y \mid y \geq 0 \}$$

- The **graph** looks like a ∇ ; The **turning point** is called the vertex.

GRAPHING
(by Table)

To create a table of values, it's best to place the vertex in the middle, then include values on both sides.

- Find the **x-value of the vertex**. Set the expression from the inside of the absolute value bars equal to 0 and solve.
- PLACE THIS VALUE IN THE MIDDLE ROW OF YOUR TABLE.** Number up and down, then complete the table.
- GRAPH!**

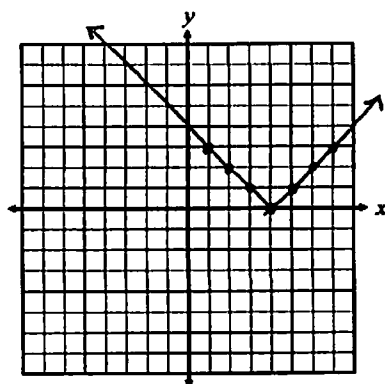
Directions: Graph each equation. Give the domain and range.

1. $f(x) = |x - 4|$

$$x - 4 = 0$$

$$x = 4$$

x	$f(x)$
1	3
2	2
3	1
4	0
5	1
6	2
7	3



$D = \{ \mathbb{R} \}$

$R = \{ y \mid y \geq 0 \}$

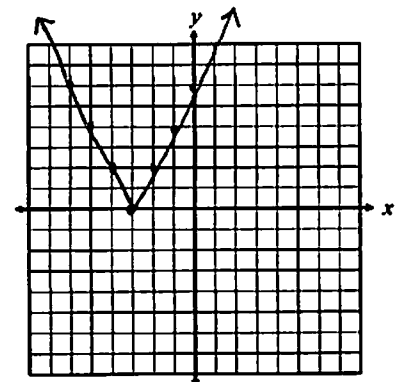
2. $f(x) = |2x + 6|$

$$2x + 6 = 0$$

$$2x = -6$$

$$x = -3$$

x	$f(x)$
-6	6
-5	4
-4	2
-3	0
-2	2
-1	4
0	6

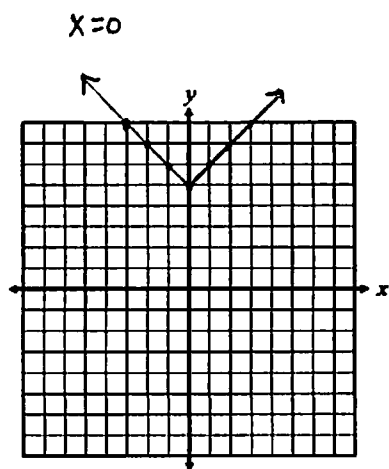


$D = \{ \mathbb{R} \}$

$R = \{ y \mid y \geq 0 \}$

3. $f(x) = |x| + 5$

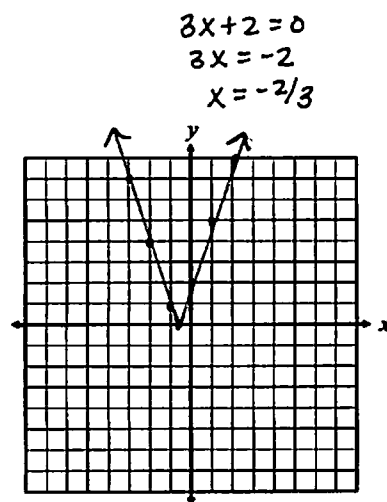
x	f(x)
-3	8
-2	7
-1	6
0	5
1	6
2	7
3	8



$D = \mathbb{R}$ $R = \{y \mid y \geq 5\}$

4. $f(x) = |3x + 2|$

x	f(x)
-3	7
-2	4
-1	1
$-2/3$	0
0	2
1	5
2	8

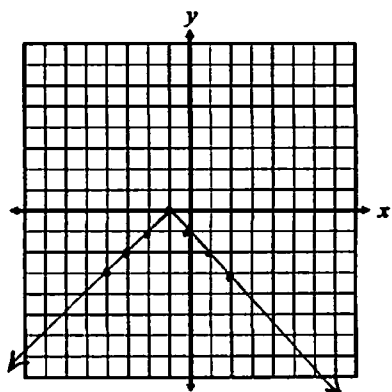


$D = \mathbb{R}$ $R = \{y \mid y \geq 0\}$

5. $f(x) = -|x + 1|$

$x + 1 = 0$
 $x = -1$

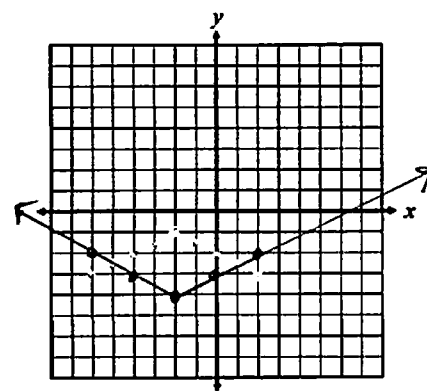
x	f(x)
-4	-3
-3	-2
-2	-1
-1	0
0	-1
1	-2
2	-3



$D = \mathbb{R}$ $R = \{y \mid y \leq 0\}$

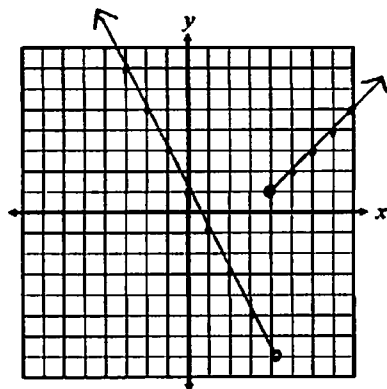
6. $f(x) = \left| \frac{1}{2}x + 1 \right| - 4$

x	f(x)
-5	-2.5
-4	-3
-3	-3.5
-2	-4
-1	-3.5
0	-3
1	-2.5



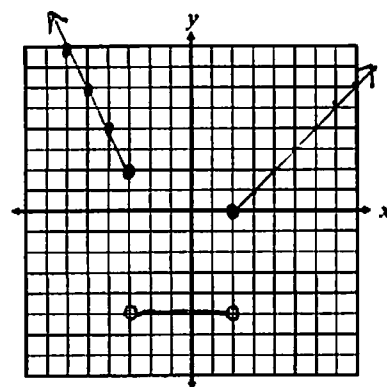
$D = \mathbb{R}$ $R = \{y \mid y \geq -4\}$

7. $f(x) = \begin{cases} -2x + 1 & \text{if } x < 4 \\ |x - 3| & \text{if } x \geq 4 \end{cases}$



$D = \mathbb{R}$ $R = \{y \mid y > -7\}$

8. $f(x) = \begin{cases} |2x| - 4 & \text{if } x \leq -3 \\ -5 & \text{if } -3 < x < 2 \\ x - 2 & \text{if } x \geq 2 \end{cases}$



$D = \mathbb{R}$ $R = \{y \mid y = -5 \cup y \geq 0\}$